

UNIT 4 – STATISTICAL INFERENCE

CO4-AT3 – VIVA VOCE ASSESSMENT

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Subject	Statistical Inference	Assessment	Viva Voce

A. Introduction to Statistical Inference

1. What is statistical inference?

Answer: Statistical inference is the process of using information obtained from a sample to draw conclusions about a population. It includes estimating unknown population quantities and making decisions or conclusions based on sample evidence.

2. Why is statistical inference important in data science?

Answer: Statistical inference is important because data scientists usually work with samples rather than complete populations. It helps them estimate population characteristics, quantify uncertainty, and make evidence-based conclusions from observed data.

3. What is the difference between descriptive statistics and inferential statistics?

Answer: Descriptive statistics summarize and describe the data that have been collected, whereas inferential statistics use sample data to draw conclusions about a larger population. Inference therefore goes beyond simply describing the observed sample.

4. What is a population in statistics?

Answer: A population is the complete set of individuals, observations, or measurements about which a statistical study is concerned. It is the entire group from which a sample may be selected.

5. What is a sample?

Answer: A sample is a subset of observations selected from a population. Statistical analysis of the sample is used to learn about the population.

6. Why do data scientists usually work with samples instead of complete populations?

Answer: Studying an entire population can be costly, time-consuming, or impractical. A representative sample can provide useful information about the population while requiring fewer observations and resources.

7. What is a population parameter?

Answer: A population parameter is a numerical characteristic of the entire population, such as the population mean, population proportion, or population variance. It is usually fixed but may be unknown.

8. What is a sample statistic?

Answer: A sample statistic is a numerical value calculated from sample observations, such as the sample mean, sample proportion, or sample variance. It is used to provide information about a population parameter.

9. Give a real-world example where statistical inference is used in industry.

Answer: For example, a manufacturer may inspect a sample of products from a production line and use the sample defect rate to infer the likely defect rate of the entire production process.

10. What are the two major activities involved in statistical inference?

Answer: The two major activities are estimation and hypothesis testing. Estimation uses sample data to estimate unknown population parameters, while hypothesis testing uses sample evidence to evaluate claims about population parameters.

B. Frequentist Approach

11. What is the frequentist approach to statistical inference?

Answer: The frequentist approach treats population parameters as fixed quantities and the observed data or samples as random. Inference is based on the behavior of statistics under repeated sampling.

12. What is the basic idea behind frequentist statistics?

Answer: The basic idea is to use repeated-sampling behavior to make conclusions about unknown population parameters. Sample statistics are studied through their sampling distributions to support estimation and hypothesis testing.

13. How does the frequentist approach differ from the Bayesian approach?

Answer: In the frequentist approach, the parameter is treated as fixed and probability describes the random behavior of data. In the Bayesian approach, probability is used to represent uncertainty about the parameter and prior information can be combined with observed data to obtain a posterior distribution.

14. In the frequentist approach, what is considered fixed: the parameter or the data?

Answer: The population parameter is considered fixed, while the data are considered random because different random samples can produce different observed values.

15. What does long-run frequency mean in frequentist statistics?

Answer: Long-run frequency is the proportion of times an event would occur if the same random procedure were repeated a very large number of times under the same conditions. Frequentist probability is interpreted through this repeated-trial behavior.

16. Why is random sampling important in the frequentist approach?

Answer: Random sampling gives the sampling process a well-defined probability behavior. This makes it possible to study how an estimator behaves across repeated samples and to quantify uncertainty in statistical inference.

17. Can a population parameter be considered a random variable in the frequentist approach?

Why?

Answer: No. In the frequentist approach, a population parameter is treated as a fixed but generally unknown quantity. Randomness is attributed to the sample and the statistics calculated from it.

18. Give an example of a frequentist problem in a data science application.

Answer: A data scientist may take a random sample of customer transactions and estimate the population average transaction value. The sample mean is treated as a random statistic whose sampling behavior is used to infer the fixed population mean.

C. Point Estimation

19. What is a point estimate?

Answer: A point estimate is a single numerical value calculated from sample data and used as an estimate of an unknown population parameter. For example, the sample mean is a point estimate of the population mean.

20. What is an estimator?

Answer: An estimator is a rule or statistical procedure that uses sample data to estimate a population parameter. For example, the sample mean is an estimator of the population mean.

21. What is the difference between an estimator and an estimate?

Answer: An estimator is the rule or formula used to obtain an estimate, whereas an estimate is the particular numerical value produced by applying that estimator to a specific sample.

22. What is the point estimator for the population mean?

Answer: The sample mean is the standard point estimator for the population mean. It is calculated as the sum of all sample observations divided by the sample size.

23. How do you estimate a population proportion from a sample?

Answer: The population proportion is estimated using the sample proportion. It is obtained by dividing the number of observations having the characteristic of interest by the total sample size.

24. What is the point estimate of population variance?

Answer: The sample variance is used as the point estimate of the population variance. The usual unbiased sample variance uses the squared deviations from the sample mean and divides their sum by $n-1$.

25. Why is the sample mean commonly used to estimate the population mean?

Answer: The sample mean uses all observations in the sample and, under standard random-sampling conditions, is an unbiased estimator of the population mean. It is therefore a natural and widely used estimator.

26. What properties should a good estimator have?

Answer: A good estimator should preferably be unbiased, have low variance, and be consistent as the sample size increases. These properties indicate that the estimator is centered appropriately, reasonably stable, and increasingly close to the true parameter.

27. What is an unbiased estimator?

Answer: An estimator is unbiased if its expected value equals the true population parameter it is intended to estimate. Thus, over repeated samples, it does not systematically overestimate or underestimate the parameter.

28. What is meant by the bias of an estimator?

Answer: Bias is the difference between the expected value of an estimator and the true population parameter. An estimator with zero bias is called unbiased.

29. What is the difference between bias and variance of an estimator?

Answer: Bias measures systematic difference between the estimator's expected value and the true parameter. Variance measures how much the estimator varies from sample to sample around its expected value. An estimator can have low bias but high variance, or vice versa.

30. Why can two different samples produce different point estimates for the same population parameter?

Answer: Different random samples contain different observations. Therefore, statistics calculated from those samples, such as the sample mean, can differ even though both samples come from the same population. This sampling variability creates uncertainty in point estimates.

D. Variability in Estimates

31. Why does a point estimate have uncertainty?

Answer: A point estimate is based on a sample rather than the complete population. Since another random sample could produce a different estimate, the point estimate has sampling uncertainty.

32. What is the sampling distribution of an estimator?

Answer: The sampling distribution of an estimator is the probability distribution of the estimator's values obtained from repeated samples of the same size from the population. It describes how the estimator varies from sample to sample.

33. What is the standard error?

Answer: Standard error measures the typical variability of a statistic or estimator across repeated samples. For a sample mean, the standard error is the standard deviation of the sampling distribution of the sample mean.

34. How is standard error different from standard deviation?

Answer: Standard deviation describes the variability of individual observations within a dataset or population. Standard error describes the variability of a statistic, such as a sample mean, across repeated samples.

35. How does increasing the sample size affect the standard error of the sample mean?

Answer: Increasing the sample size generally decreases the standard error of the sample mean. For independent observations with finite variance, the standard error is proportional to 1 divided by the square root of the sample size.

36. Why does variability in an estimate decrease when the sample size increases?

Answer: A larger sample contains more information about the population. Random fluctuations tend to average out more effectively, so the estimator becomes more stable and its sampling variability decreases.

37. What is the relationship between standard error and confidence intervals?

Answer: Standard error measures the sampling variability of an estimator and is a key component in constructing a confidence interval. A larger standard error generally produces a wider interval, while a smaller standard error produces a narrower interval, when other factors remain the same.

38. If two samples have different sample means, does it necessarily mean that one sample is incorrect? Explain.

Answer: No. Different random samples can naturally have different sample means because of sampling variability. Both samples may be valid and representative of the same population even though their calculated means are different.

E. Confidence Intervals

39. What is a confidence interval?

Answer: A confidence interval is an interval calculated from sample data that provides a range of plausible values for an unknown population parameter, together with a specified confidence level.

40. Why is a confidence interval more informative than a point estimate?

Answer: A point estimate gives only one value, while a confidence interval provides a range that reflects sampling uncertainty. It therefore communicates both an estimated value and the uncertainty associated with that estimate.

41. What does a 95% confidence level mean in the frequentist approach?

Answer: It means that if the same sampling and interval-construction procedure were repeated many times, approximately 95% of the resulting intervals would contain the true population parameter. It does not mean that there is a 95% probability that one already-computed fixed interval contains the parameter.

42. What is the difference between a confidence level and a confidence interval?

Answer: The confidence level is the long-run coverage rate of the interval procedure, such as 95%. The confidence interval is the actual range of values calculated from a particular sample.

43. What factors determine the width of a confidence interval?

Answer: The width depends mainly on the variability of the data or standard error, the sample size, and the selected confidence level. Greater variability and higher confidence levels tend to widen the interval, while larger sample sizes tend to narrow it.

44. How does increasing the sample size affect the width of a confidence interval?

Answer: Increasing the sample size generally reduces the standard error, so the confidence interval becomes narrower when the confidence level and other conditions remain unchanged.

45. How does increasing the confidence level from 95% to 99% affect the confidence interval?

Answer: Increasing the confidence level generally makes the interval wider because a higher level of confidence requires a larger margin around the point estimate.

46. What is the difference between a z-confidence interval and a t-confidence interval?

Answer: A z-confidence interval uses a critical value from the standard normal distribution, while a t-confidence interval uses a critical value from the t-distribution. The t-distribution accounts for additional uncertainty when the population standard deviation is estimated from the sample, especially for smaller samples.

47. When would you use the t-distribution instead of the normal distribution for estimating a population mean?

Answer: The t-distribution is commonly used when estimating a population mean when the population standard deviation is unknown and is estimated using the sample standard deviation, particularly when the sample size is small and the population is reasonably close to normal.

F. Hypothesis Testing Using Confidence Intervals

48. What is hypothesis testing?

Answer: Hypothesis testing is a statistical procedure used to evaluate a claim about a population parameter using sample data. It compares the observed evidence with what would be expected under a null hypothesis.

49. What are the null hypothesis (H_0) and alternative hypothesis (H_1)?

Answer: The null hypothesis, H_0 , is the initial claim or reference assumption being tested. The alternative hypothesis, H_1 , represents the competing claim that is considered when the sample provides sufficient evidence against H_0 .

50. How can a confidence interval be used to make a decision about a hypothesis?

Answer: A confidence interval can be compared with the value specified by the null hypothesis. For a corresponding two-sided test, if the null value lies outside the confidence interval, the null hypothesis is rejected at the associated significance level; if it lies inside, there is not sufficient evidence to reject the null hypothesis.

Assessment Criteria

- Conceptual understanding of statistical inference.
- Accuracy in explaining statistical terminology and principles.
- Ability to distinguish population, sample, parameter, statistic, estimator, and estimate.
- Understanding of sampling variability, standard error, and confidence intervals.
- Ability to explain the frequentist approach and hypothesis testing clearly.
- Clarity, confidence, and logical organization of the viva responses.